# Felix Technical Report: Balanced-Ternary Discrepancy Bounds, SPQT Base-243 Packing, and SPARK Formal Verification

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## 1. Discrete Mathematics and Graph Percolation Verifiers

Module: crates/felix-lang/src/erdos_pqt_omniversal.rs | SPARK: ada/erdos_pqt_spark.ads

- Geometrically Damped Ternary Stride Sum: D(d) = |sum tau(k*d) * 2^(-k)| <= 1.0
  (Evaluated across first 16 primes: max_disc = 0.285714 <= 1.0; stride 3 = 0.0)
- Erdos-Renyi Percolation Threshold: p_c = ln(N)/N = 0.0004063 (for N = 24,917)
- Exhaustive Ramsey R(3,3) = 6 Check: All 2^15 = 32,768 2-colorings of K_6 verified
- Erdos-Kac Distinct Prime Factor Count omega(n) & Sunflower Bitwise Intersection

## 2. Zero-Allocation Signal Resolution & 3-Byte SPQT Base-243 Packing

Modules: well_signal_resolution.rs, well_signal_spqt_espnow.rs, well_signal_spqt_spark.ads

- 5-Trit Bijection: B = sum_{j=0..4} (t_j + 1) * 3^j in [0, 242]
- Lap 1 (Un-bended Container): 6/12 trits -> SPQT3 = [239, 205, 238] (t_14 = +1)
- Lap 2 (Overruled by Later Lap): 9/12 trits -> SPQT3 = [239, 233, 104] (t_14 = 0)
- StackMarkdownBuf: 444 / 512 bytes rendered with 0 heap allocations

## 3. OS-Agnostic Capability ABI, Webhook Verifier & 6-Step Q_6 Inference

Modules: phoenix_cap_abi.rs, m8_axum_webhook_server.rs, self_other_hypercube_inference.rs

- Hardware-enforced capability attenuation across POSIX, WASM, and ESP32-S3 backends
- Constant-time RFC 2104 HMAC-SHA256 webhook verification & one-time epoch tickets
- 6-Step Q_6 Statistical Inference: coherence = 0.9705, Pearson r = 0.9999, do = 1.1200

## 4. Contributor Table & Rational p/q Signature Map

| Contributor / Handle | Role & Subsystem | p/q Signature |
|---|---|---|
| Peter Lodri (peterlodri-sec) | Core Architecture, Rust & SPARK | 31/32 |
| Alexandra (ALEX\|qMom) | Project Stewardship & Review | 528/432 = 11/9 |
| Chris | Hardware & Acoustic Systems | 288/243 = 32/27 |
| Dra | Operations & Executive Review | 144/89 |
| Nate / Natalie (standardgalactic) | memnet Oracle & Corpus Spine | 243/256 |

### 5. Embedded Verification & Rational p/q Signature

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Combined Rational Product : P/Q = 44599 / 1424 (irreducible)
LaTeX Source SHA-256      : fecceb37c83c9da0013d489458d1c7e73153b2d679834cba7eb84663b7375530
Verification Status       : 106/106 Rust tests PASS | 22 SPARK packages proved
```